

Mathematics

Inline math sits on the baseline of the text: the roots of $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, and Euler's identity $e^{i\pi} + 1 = 0$ links five constants.

Display equations

The Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

A sum and a limit:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

Aligned equations

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0$$

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Matrices and cases

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Math in a list

2. The integral of $\frac{1}{x}$ is $\ln|x| + C$.

An error

A typo in LaTeX should show in red without breaking the rest of the note: `\frac{1}{`.